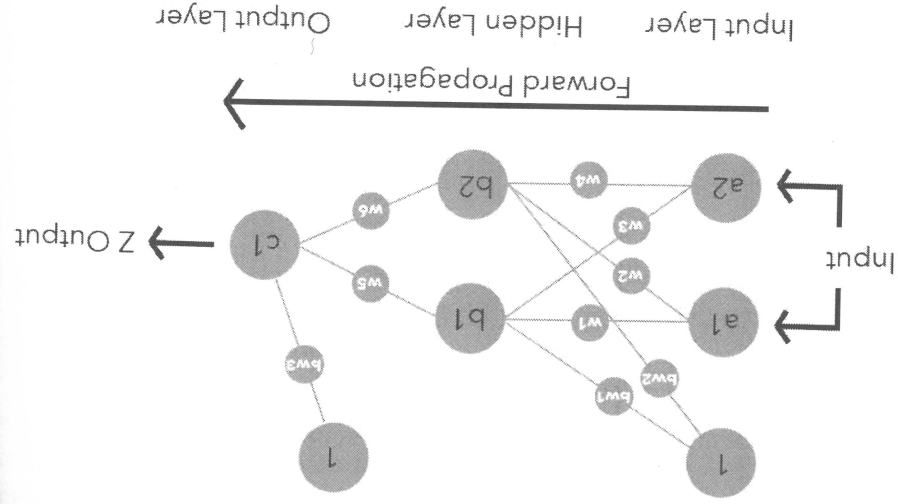


## Stage 1: Forward Propagation

Now that we understand the structure of a neural network, we can investigate how input *moves through the network to become output*. This process is technically called *forward propagation*.



When input is passed into the network, it moves from one layer to the next until it passes through the output layer. A total of two mathematical functions are repeatedly used to make all of this possible, and we will examine both in this chapter.

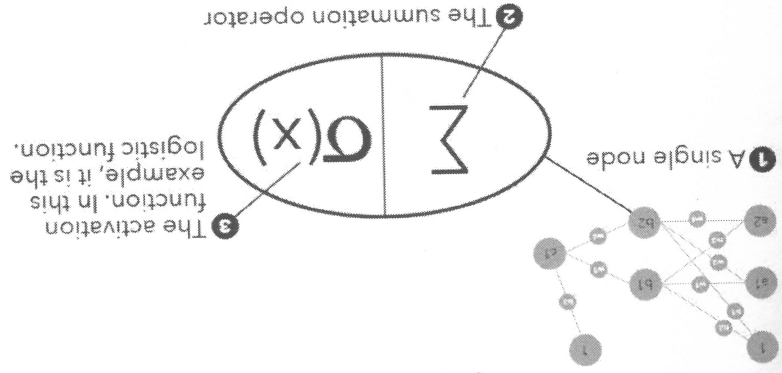
- This stage is organized into three parts.
1. Understanding The Mathematical Functions Used
  2. Understanding Matrices
  3. Fitting it All Together: Review

## Ch. 5 Understanding the Mathematical Functions Used

There are a total of two mathematical functions used in this stage, and both of these functions occur *inside of every hidden and output node*. Yes, you heard right: every single node! These functions include:

- Summation operator
- Activation function

A fantastic way to understand how and where these functions operate is to view them as a *working pair* within each node:



### What is a Summation Operator?

The summation operator sums a sequence of numbers, and is a convenient tool to use when dealing with large number sets and/or the repetition of an operation (such as addition or multiplication). Within a neural network, the summation operator sums all of a node's inputs to create a *net input*. Below is the operator in its general form, followed by its specific usage for calculating the net input of a node.